YEAR 10 MATHS · CHAPTER 7 · MIND MAP

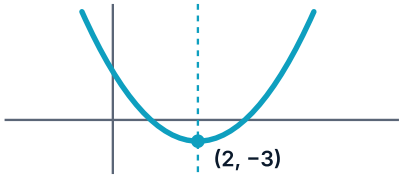
Parabolas and rates of change

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1 Exploring parabolas

$y = ax^2 + bx + c$: axis of symmetry $x = -\frac{b}{2a}$, y-intercept c.

$y = x^2 - 4x + 1$: axis $x = 2$, turning point $(2, -3)$



2 Sketching with transformations

$y = a(x - h)^2 + k$ has turning point (h, k).

$y = -2(x + 1)^2 + 3$: upside down, narrower, turning point $(-1, 3)$

3 Sketching using factorisation

Factorise for the x-intercepts. The turning point is halfway between them.

$y = x^2 - 2x - 8 = (x - 4)(x + 2)$

x-intercepts −2 and 4, axis $x = 1$, turning point $(1, -9)$

4 Sketching by completing the square

Complete the square to read off the turning point.

$y = x^2 + 6x + 5 = (x + 3)^2 - 4$

Turning point $(-3, -4)$

5 The quadratic formula and discriminant

$\Delta = b^2 - 4ac$ tells you how many x-intercepts.

$\Delta > 0$: two. $\Delta = 0$: touches once. $\Delta < 0$: none

$y = x^2 + 2x + 5$: $\Delta = 4 - 20 < 0$, no x-intercepts

6 Applications of parabolas

The maximum or minimum is at the turning point.

$h = -5t^2 + 20t + 1$: $t = 2$, max height 21 m

40 m of fence: $A = x(20 - x)$, biggest at $x = 10$: 100 m²

7 Lines and parabolas

Set the equations equal and solve. The discriminant tells you how many points they share.

$y = x^2$ and $y = x + 2$: $x^2 - x - 2 = 0$

$x = 2$ or -1: points $(2, 4)$ and $(-1, 1)$

8 Rates of change

Rate $= \frac{\text{change in } y}{\text{change in } x}$, in units per unit.

300 km in 4 hours: **75 km/h**

The gradient of a distance–time graph is the speed

9 Average and instantaneous rates

Average: gradient of the chord between two points. Instantaneous: gradient of the tangent at one point.

$y = x^2$ from $x = 1$ to 3: $\frac{9-1}{3-1} = 4$

At $x = 2$, using $x = 2.01$: $\frac{4.0401-4}{0.01} = 4.01 \approx 4$

10 Direct and inverse variation

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

$y \propto x$, $y = 12$ when $x = 3$: $k = 4$, so at $x = 5$, $y = 20$

$y \propto \frac{1}{x}$, $y = 6$ when $x = 2$: $k = 12$, so at $x = 4$, $y = 3$